EXPERIMENT 5: MISSIONARIES AND CANNIBALS PROBLEM

Aim

To write a Python program to solve the **Missionaries and Cannibals Problem** using Artificial Intelligence techniques.

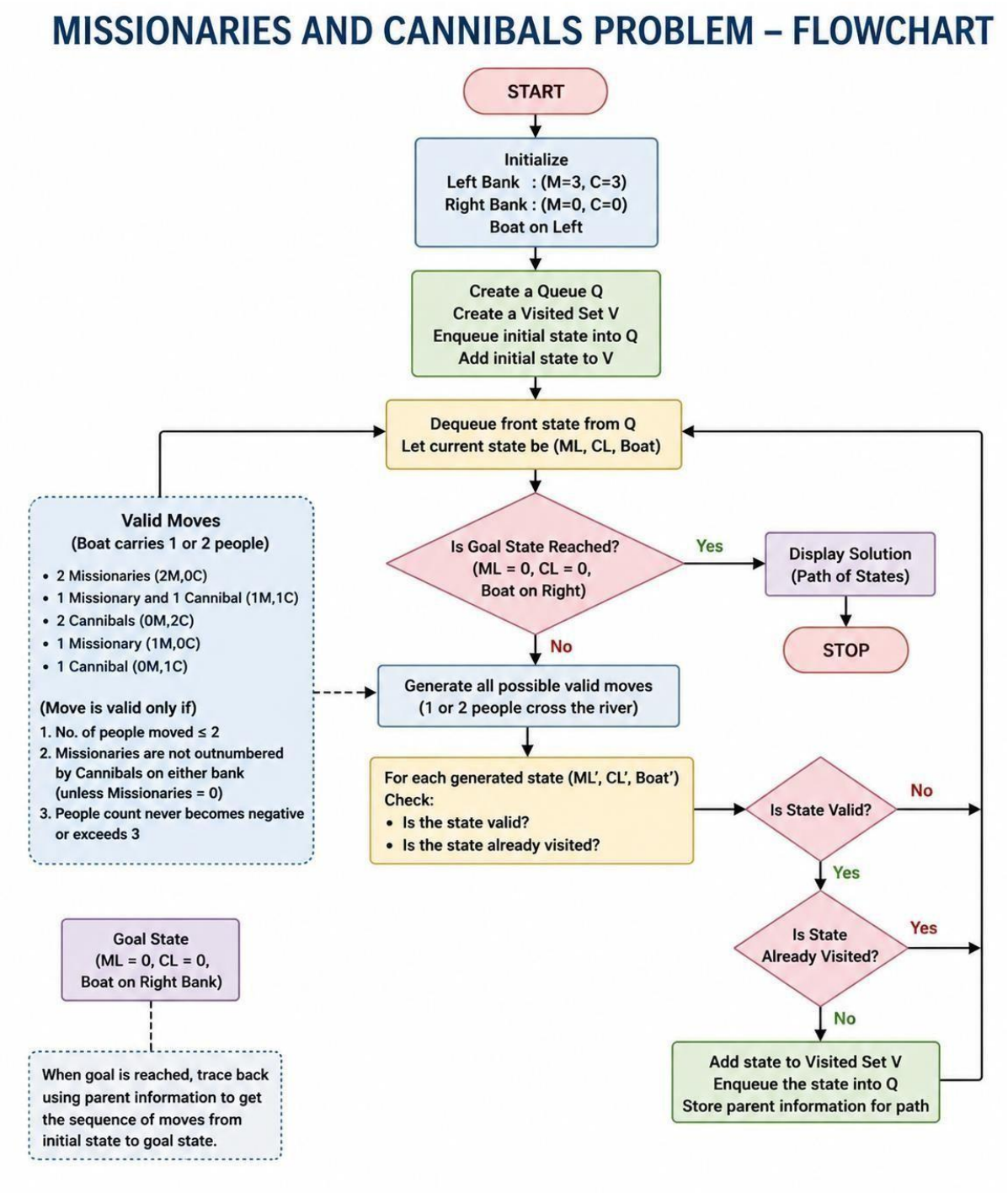
Objective

- To understand state-space search techniques in Artificial Intelligence.
- To represent a problem using states and operators.
- To find a valid sequence of moves to reach the goal state.

Algorithm

1. Start.
2. Define the initial state as:
    - 3 Missionaries on the left side. ○ 3 Cannibals on the left side.
    - Boat on the left side.
3. Define the goal state as:
    - 0 Missionaries and 0 Cannibals on the left side.
4. Generate all possible boat moves.
5. Check whether each generated state is valid.
6. Avoid already visited states.
7. Continue searching until the goal state is reached.
8. Display the sequence of steps.
9. Stop.

Flowchart:



Code (Python) from

collections import deque

def is_valid(m, c):

Missionaries should not be outnumbered return

(m == 0 or m >= c) and (3-m == 0 or 3-m >= 3-c) def

missionaries_cannibals():

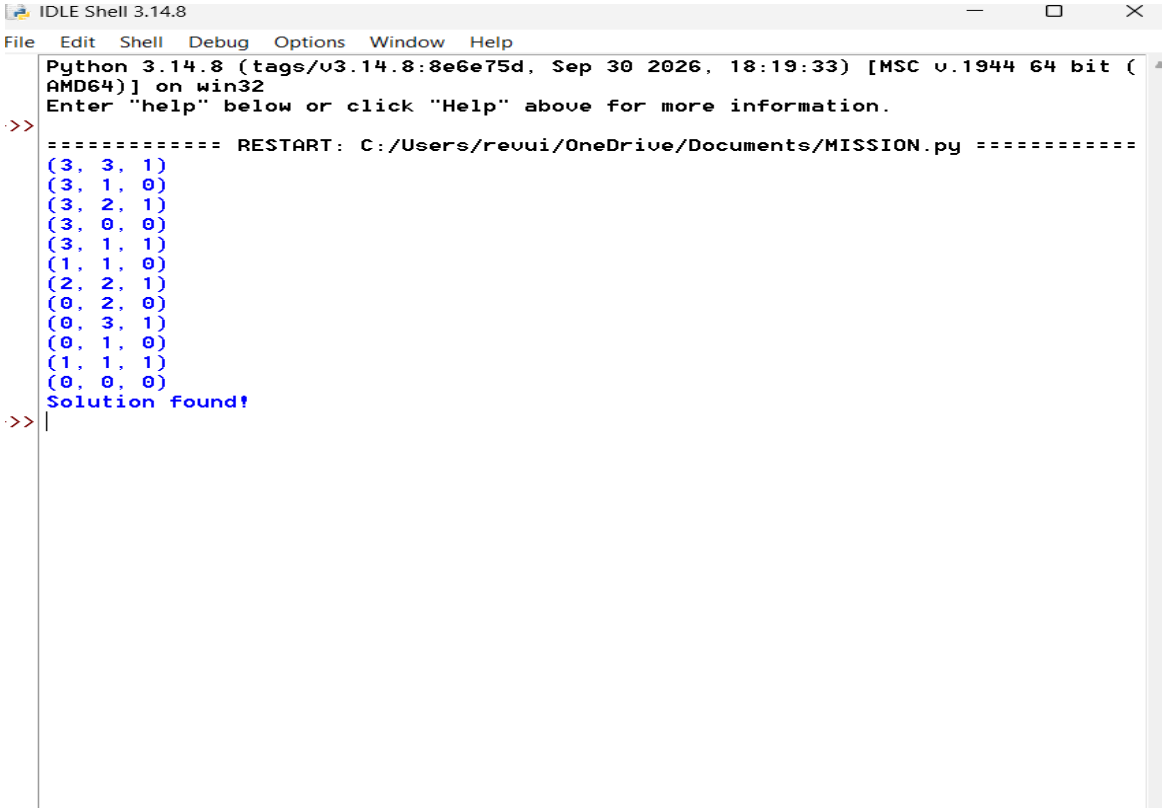
```
    start = (3, 3, 1)
goal = (0, 0, 0)
    queue = deque([(start, [])])
visited = set()
    while queue:
        state, path = queue.popleft()
        if state in visited:
continue
        visited.add(state)
path = path + [state]
        if state == goal:
return path
        m, c, boat = state
        moves = [
            (1,0), (2,0),
            (0,1), (0,2),
            (1,1)
        ]
        for dm, dc in moves:
            if boat == 1:
                new_state = (m-dm, c-dc, 0)
            else:
                new_state = (m+dm, c+dc, 1)
            if (0 <= new_state[0] <= 3 and                0
<=          new_state[1]          <=          3          and
is_valid(new_state[0], new_state[1])):
                queue.append((new_state, path))
solution = missionaries_cannibals()
```

print("Solution Path:") for

step in solution:

print(step)

Output



Result

Thus, the Python program to solve the **Missionaries and Cannibals Problem** was executed successfully, and the sequence of valid moves to reach the goal state was obtained.

Conclusion

The experiment helped in understanding state-space representation and search algorithms in Artificial Intelligence. The Missionaries and Cannibals Problem was successfully solved by exploring possible states and selecting valid transitions.